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JOINT STRUCTURE

Abstract

A joint structure includes an upper block arranged on a lower block, a sleeve inserted into the upper block, an auxiliary member provided on an upper surface of the upper block, a pin in which a first shaft portion is inserted into a first pin insertion slit and a second pin insertion slit, and a first male threaded portion is screwed into the lower block, and a bolt. The first pin insertion slit of the upper block opens to one side surface of the upper block in a longitudinal direction, the second pin insertion slit of the auxiliary member opens to one side surface of the auxiliary member in the longitudinal direction. A lower surface of the auxiliary member is formed by a pair of inclined surfaces that gradually extends away from an upper surface of the auxiliary member from both ends in the longitudinal direction toward a center.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation under 35 U.S.C. § 120 of No. PCT/JP2023/037694, filed Oct. 18, 2023, which is incorporated herein by reference, and which claimed priority to Japanese Application No. 2022-187996, filed Nov. 25, 2022. The present application likewise claims priority under 35 U.S.C. § 119 to Japanese Application No. 2022-187996, filed Nov. 25, 2022, the entire content of which is also incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a joint structure for use in semiconductor manufacturing equipment and the like.

BACKGROUND

[0003] Japanese Patent Application Publication No. 2005-164021 discloses a joining device for joining an upper block and a lower block. In the joining device of Patent Literature 1, a positioning pin is fixed to the lower block and inserted from above into a pin insertion hole of the upper block, and the upper block and the lower block are overlapped. Next, a notch of an auxiliary member is fitted into the small diameter portion of the positioning pin from the lateral direction. Then, a bolt is inserted into each bolt insertion hole of a tightening auxiliary member and the upper block and screwed into a screw hole of the lower block. In this way, the upper block and the lower block are joined.

SUMMARY

[0004] However, in the joining device of Japanese Patent Application Publication No. 2005-164021, when a sleeve having a fluid passage inside is provided between the pin insertion hole and the bolt insertion hole in the upper block, it is difficult to attach the auxiliary member in the above-mentioned manner.

[0005] Therefore, one of the objects of the present disclosure is to provide a joint structure that allows an upper block provided with a sleeve to be easily assembled to a lower block.

[0006] A joint structure according to one embodiment of the present disclosure includes an upper block arranged on a lower block having a fluid passage, in which a first pin insertion slit, a first sleeve insertion hole, and a first bolt insertion hole are formed, and the first sleeve insertion hole is located between the first pin insertion slit and the first bolt insertion hole; an auxiliary member provided on an upper surface of the upper block, in which a second pin insertion slit, a second sleeve insertion hole, and a second bolt insertion hole are formed, and the second sleeve insertion hole is located between the second pin insertion slit and the second bolt insertion hole; a sleeve inserted into the first sleeve insertion hole of the upper block and the second sleeve insertion hole of the auxiliary member, an inside of which serves as a fluid passage communicating with the fluid passage of the lower block; a pin having a first head, a first shaft portion, and a first male threaded portion, the first head abutting against an upper surface of the auxiliary member, the first shaft portion being inserted into the first pin insertion slit and the second pin insertion slit, and the first male threaded portion being screwed into the lower block; and a bolt having a second head, a second shaft portion, and a second male threaded portion, the second head abutting against the upper surface of the auxiliary member, the second shaft portion being inserted into the first bolt

insertion hole and the second bolt insertion hole, and the second male threaded portion being screwed into the lower block. The first pin insertion slit opens to one side surface of the upper block in a longitudinal direction. The second pin insertion slit opens to one side surface of the auxiliary member in the longitudinal direction. A lower surface of the auxiliary member is formed by a pair of inclined surfaces that gradually extends away from the upper surface of the auxiliary member from both ends in the longitudinal direction toward a center.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In the drawings:

[0008] FIG. 1 is a cross-sectional view of a joint structure according to a first embodiment;

[0009] FIG. 2A is a plan view and FIG. 2B is a cross-sectional view of an upper block;

[0010] FIG. 3 is a side view of a pin;

[0011] FIG. 4A is a plan view and FIG. 4B is a cross-sectional view of an auxiliary member;

[0012] FIG. 5A to FIG. 5D are explanatory diagrams of a method for assembling the joint structure;

[0013] FIG. 6A to FIG. 6C are explanatory diagrams of a method for assembling the joint structure; and

[0014] FIG. 7 is a cross-sectional view of a joint structure according to a second embodiment.

DETAILED DESCRIPTION

[0015] A joint structure according to one embodiment of the present disclosure will be described with reference to the drawings. The joint structure constitutes a part of a gas line of an integrated type fluid control device.

[0016] FIG. 1 is a cross-sectional view of a joint structure 1 according to a first embodiment.

[0017] The joint structure 1 includes a lower block 10, an upper block 20, a sleeve 30, a gasket portion 40, a pin 50, a bolt 60, and an auxiliary member 70.

[0018] The lower block 10 has a substantially rectangular parallelepiped shape and has a fluid passage 11 that is bent at a right angle. The lower block 10 has an upper surface 12. The fluid passage 11 opens at a central portion of the upper surface 12. The lower block 10 is formed with a pin screw hole 13 and a bolt screw hole 14, which are arranged on either side of the fluid passage 11, and both holes open to the upper surface 12. The upper surface 12 has a first annular recess 15 formed around the opening of the fluid passage 11.

[0019] FIG. 2A shows a plan view and FIG. 2B shows a cross-sectional view of the upper block 20.

[0020] As shown in FIG. 1, the upper block 20 has a substantially rectangular parallelepiped shape and is disposed on the upper surface 12 of the lower block 10. As shown in FIG. 2A and FIG. 2B, the upper block 20 has a first pin insertion slit 21, a first sleeve insertion hole 22, and a first bolt insertion hole 23. In a longitudinal direction of the upper block 20, the first sleeve insertion hole 22 is located at the center, and the first pin insertion slit 21 and the first bolt insertion hole 23 are located on both sides of the first sleeve insertion hole 22. The first pin insertion slit 21 opens to an upper surface 24 and a lower surface 25 of the upper block 20, and also opens to one side surface 26 of the upper block 20 in the longitudinal direction. The first pin insertion slit 21 has a semi-oval shape (semi-elongated circular shape or semi-elliptical shape) in plan view.

[0021] The first sleeve insertion hole 22 opens to the upper surface 24 and the lower surface 25 of the upper block 20. A pair of first notches 22a is formed on the upper surface 24 of the upper block 20. Each of the first notches 22a extends from the first sleeve insertion hole 22 to penetrate a side surface 27 parallel to the longitudinal direction. Each of the first notches 22a has a rectangular shape when viewed from the side surface 27 side. The lower surface 25 of the upper block 20 has a second annular recess 28 formed around the opening of the first sleeve insertion hole 22. A pair of

second notches **29** is formed on the lower surface **25** of the upper block **20**. Each second notch **29** extends from the first sleeve insertion hole **22** to penetrate the side surface **27** parallel to the longitudinal direction. Each second notch **29** has a rectangular shape when viewed from the side surface **27** side. The first bolt insertion hole **23** opens to the upper surface **24** and the lower surface **25** of the upper block **20**. The first sleeve insertion hole **22** and the first bolt insertion hole **23** have a substantially circular shape in plan view.

[0022] As shown in FIG. **1**, the sleeve **30** is cylindrical, and the inside thereof serves as a fluid passage. This fluid passage communicates with the fluid passage **11** of the lower block **10**. The sleeve **30** has two straight portions **31** and **32** and a bent portion **33**. The straight portion **31** extends in a direction perpendicular to the upper surface **24** of the upper block **20**, and the straight portion **32** extends parallel to the upper surface **24** of the upper block **20**. The two straight portions **31** and **32** are connected by the bent portion **33**. A lower end portion of the straight portion **31** is inserted into the first sleeve insertion hole **22**. A flange portion **34** is provided at the lower end of the straight portion **31**. The flange portion **34** is fitted into the second recess **28**. An annular third recess **35** is formed on a lower surface of the straight portion **31** and the flange portion **34**. The straight portion **32** and the bent portion **33** are positioned above the pin **50** so as to cover the pin **50**.

[0023] As shown in FIG. **1**, the gasket portion **40** includes a gasket **41**, a retainer **42**, and a snap ring **43**. The gasket **41** and the retainer **42** are annular in shape. The gasket **41** is held by the retainer **42**. The retainer **42** is held by the snap ring **43** formed of a metal wire having a circular cross section and formed into a C-shape. The gasket **41**, the retainer **42**, and the snap ring **43** are accommodated in the first recess **15** and the third recess **35**.

[0024] FIG. **3** shows a side view of the pin **50**.

[0025] As shown in FIG. **1** and FIG. **3**, the pin **50** includes a first head **51**, a first shaft portion **52**, and a first male threaded portion **53**. The first head **51** is disk-shaped. The first shaft portion **52** extends downward from a lower surface **51A** of the first head **51** and is inserted into the first pin insertion slit **21**. The first shaft portion **52** has a pair of flat surfaces **52A** at its upper end. The first male threaded portion **53** is provided at a lower end portion of the first shaft portion **52** and is screwed into the pin screw hole **13**. An outer diameter of the first male threaded portion **53** is configured to be smaller than an outer diameter of the first shaft portion **52**. Therefore, a step portion **54** is formed by the first shaft portion **52** and the first male threaded portion **53**. A step surface **55** of the step portion **54** abuts against the upper surface **12** of the lower block **10**.

[0026] As shown in FIG. **1**, the bolt **60** includes a second head **61**, a second shaft portion **62**, and a second male threaded portion **63**. A hexagonal hole is formed in the head **61**. The second shaft portion **62** extends downward from the second head **61** and is inserted into the first bolt insertion hole **23**. The second male threaded portion **63** is provided at a lower end of the second shaft portion **62** and is screwed into the bolt screw hole **14**.

[0027] FIG. **4A** shows a plan view and FIG. **4B** shows a cross-sectional view of the auxiliary member **70**.

[0028] The auxiliary member **70** has a rectangular plate-like shape in plan view, and is disposed on the upper surface **24** of the upper block **20**. As shown in FIG. **4A** and FIG. **4B**, the auxiliary member **70** includes a second pin insertion slit **71**, a second sleeve insertion hole **72**, and a second bolt insertion hole **73**. In a longitudinal direction of the auxiliary member **70**, the second sleeve insertion hole **72** is located approximately at the center, and the second pin insertion slit **71** and the second bolt insertion hole **73** are located on both sides of the second sleeve insertion hole **72**. In the present embodiment, the second pin insertion slit **71** and the second sleeve insertion hole **72** are connected and configured as one slit. The second pin insertion slit **71** opens to an upper surface **74** and a lower surface **75** of the auxiliary member **70**, and also opens to one side surface **76** of the auxiliary member **70** in the longitudinal direction. The first pin insertion slit **71** has a semi-oval shape (semi-elongated circular shape or semi-elliptical shape) in plan view. A diameter of the arc portion of the second pin insertion slit **71** is smaller than a diameter of the arc portion of the second

sleeve insertion hole **72**, and the auxiliary member **70** has a constricted portion **77** at the connection portion between the second pin insertion slit **71** and the second sleeve insertion hole **72**. The constricted portion **77** prevents the sleeve **30** from coming out of the second sleeve insertion hole **72**.

[0029] The second sleeve insertion hole **72** opens to the upper surface **74** and the lower surface **75** of the auxiliary member **70**. The second bolt insertion hole **73** opens to the upper surface **74** and the lower surface **75** of the auxiliary member **70**. The second sleeve insertion hole **72** and the second bolt insertion hole **73** have a semi-oval shape (semi-elongated circular shape or semi-elliptical shape) in plan view. The lower surface **75** of the auxiliary member **70** is formed by a pair of inclined surfaces **77**. The pair of inclined surfaces **77** makes the auxiliary member **70** thickest at its central portion in the longitudinal direction and thinnest at its ends. That is, the auxiliary member **70** has a pair of inclined surfaces **77** that gradually extends away from the upper surface **74** from both ends in the longitudinal direction toward the center. The auxiliary member **70** has a downward protrusion **78** formed by the pair of inclined surfaces **77**.

[0030] Next, a method for assembling the joint structure **1** will be described.

[0031] FIG. 5A to FIG. 5D and FIG. 6A to FIG. 6C are explanatory diagrams of the method for assembling the joint structure **1**.

[0032] As shown in FIG. 5A, the first male threaded portion **53** of the pin **50** is screwed into the pin screw hole **13** of the lower block **10** until the step surface **55** abuts against the upper surface **12**. With an assembly jig (not shown) clamping the pair of flat surfaces **52A**, the pin **50** is screwed into the lower block **10** by rotating the pin **50**. The sleeve **30** to which the gasket portion **40** is attached is inserted into the first sleeve insertion hole **22** of the upper block **20**, and the second sleeve insertion hole **72** of the auxiliary member **70** is inserted into the sleeve **30**. Specifically, the sleeve **30** up to the straight portion **31** is inserted through the upper block **20** and the auxiliary member **70**. Then, a pipe that will become the bent portion **33** and the straight portion **32** is welded to the straight portion **31**. The long straight portion **31** may be bent to form the bent portion **33** and the straight portion **32**.

[0033] As shown in FIG. 5B, the upper block **20** and the sleeve **30** are slightly tilted, and one inclined surface **77** of the auxiliary member **70** abuts against the upper surface **24** of the upper block **20**, and then the auxiliary member **70** is inserted between the first head **51** of the pin **50** and the lower block **10**. At this time, the first shaft portion **52** of the pin **50** is inserted into the first pin insertion slit **21** and the second pin insertion slit **71**.

[0034] As shown in FIG. 5C, the gasket portion **40** is fitted into the first recess **15** of the lower block **10**. As shown in FIG. 5D, the auxiliary member **70** is pushed toward the pin **50**, and the protrusion **78** of the auxiliary member **70** faces the pair of first notches **22a**.

[0035] As shown in FIG. 6A and FIG. 6B, the bolt **60** is inserted into the second bolt insertion hole **73** and the first bolt insertion hole **23**, and the male threaded portion **63** is screwed into the bolt screw hole **14**. As a result, the upper block **20** is pushed toward the lower block **10** by the bolt **60**, the protrusion **78** of the auxiliary member **70**, and the first head **51** of the pin **50**, and the gasket **41** is deformed. As shown in FIG. 6C, when the assembly of the joint structure **1** is completed, the upper surface **74** of the auxiliary member **70** and the upper surface **24** of the upper block **20** are parallel to each other.

[0036] When the assembly of the joint structure **1** is completed, a dimension from the upper surface **12** of the lower block **10** to the lower surface **51A** of the first head **51** of the pin **50** (from the step surface **55** to the lower surface **51A**) is configured to be the same as the sum of the thickness of the upper block **20** and the thickness of the auxiliary member **70** at the protrusion **78**. In addition, the dimension from the upper surface **12** of the lower block **10** to the lower surface **51A** of the first head **51** of the pin **50** (from the step surface **55** to the lower surface **51A**) is configured to be larger than the sum of the thickness of the upper block **20** and the thickness of the end of the auxiliary member **70** on the second pin insertion slit **71** side. Therefore, the upper block **20** and the auxiliary

member **70** can be inserted between the upper surface **12** of the lower block **10** and the lower surface **51A** of the first head **51** of the pin **50**, and the gasket **41** can be deformed to obtain a desired sealing effect.

[0037] For example, the joint structure **1** of the present embodiment can be realized by using the upper block **20**, the pin **50**, and the auxiliary member **70** having the following dimensions. The thickness **X1** of the upper block **20** shown in FIG. 2A and FIG. 2B is 14.5 mm, the longitudinal width **X2** of the upper block **20** is 30.5 mm, the depth **X3** of the first pin insertion slit **21** is 8.25 mm, and the radius **R1** of the arc portion of the first pin insertion slit **21** is 2.6 mm. The diameter **Y1** of the first head **51** of the pin **50** shown in FIG. 3 is 10 mm, and the length from the lower surface **51A** of the first head **51** to the step surface **55** is 18.2 mm. The longitudinal width **Z1** of the auxiliary member **70** shown in FIG. 4A and FIG. 4B is 30.5 mm, the depth **Z2** of the second pin insertion slit **71** (length to the boundary with the second sleeve insertion hole **72**) is 6.4 mm, the longitudinal width **Z3** of the second sleeve insertion hole **72** is 12.7 mm, and the longitudinal width **Z4** of the second bolt insertion hole **73** is 6.4 mm. The radius **R2** of the arc portion of the second pin insertion slit **71** is 2.6 mm, the radius **R3** of the arc portion of the second sleeve insertion hole **72** is 3.7 mm, and the radius **R4** of the arc portion of the second bolt insertion hole **73** is 2.2 mm. The angles **θ1** and **θ2** of the inclined surfaces **77** are 5°.

[0038] According to the above joint structure **1**, the first pin insertion slit **21** opens to one side surface **26** of the upper block **20** in the longitudinal direction, the second pin insertion slit **71** opens to one side surface **76** of the auxiliary member **70** in the longitudinal direction, and the lower surface **75** of the auxiliary member **70** is formed by a pair of inclined surfaces **77** that gradually extends away from the upper surface **74** of the auxiliary member **70** from both ends in the longitudinal direction toward the center. With this configuration, the upper block **20** and the sleeve **30** can be slightly inclined, and with one inclined surface **77** of the auxiliary member **70** abutting against the upper surface **24** of the upper block **20**, they can be inserted between the first head **51** of the pin **50** and the lower block **10** from the lateral direction, and the first shaft portion **52** of the pin **50** can be inserted into the first pin insertion slit **21** and the second pin insertion slit **71**. This makes it easier to assemble the upper block **20** provided with the sleeve **30** to the lower block **10**. In addition, in a fluid control device in which pipes, such as the straight portion **32** and the bent portion **33** of the sleeve **30**, are bent and densely arranged so as to cover one side of the upper block **20**, there may be cases where the upper block **20** needs to be removed due to changes in specifications or other reasons. In such cases, there is no space for tools to enter on the side where the pipes are bent. However, with the joint structure **1**, the upper block **20** can be removed by simply removing the bolt **60** on one side. In this way, according to the joint structure **1**, the upper block **20** can be easily attached and removed using only the bolt **60** on one side.

[0039] With one inclined surface **77** of the auxiliary member **70** abutting against the upper surface **24** of the upper block **20**, the auxiliary member **70** can be pushed toward the pin **50**. This makes it possible to stabilize the posture of the auxiliary member **70** during assembly, making it easier to assemble the joint structure **1**.

[0040] The first pin insertion slit **21** and the second pin insertion slit **71** have a semi-oval shape in plan view. Therefore, the first shaft portion **52** of the pin **50** can be easily inserted into the first pin insertion slit **21** and the second pin insertion slit **71**. The second sleeve insertion hole **72** and the second bolt insertion hole **73** have an oval shape in plan view. Therefore, even when the sleeve **30** is inserted into the auxiliary member **70**, the auxiliary member **70** can be moved, so that the auxiliary member **70** can be easily moved to a predetermined position in the upper block **20**.

[0041] The upper surface **24** of the upper block **20** is formed with the first notch **22a** extending from the first sleeve insertion hole **22** to penetrate the side surface **27** parallel to the longitudinal direction of the upper block **20**. The protrusion **78** of the auxiliary member **70** fits into this first notch **22a**, so that the force with which the auxiliary member **70** presses the upper block **20** can be stabilized.

[0042] The lower surface **25** of the upper block **20** is formed with a second notch **29** extending from the first sleeve insertion hole **22** to penetrate the side surface **27** parallel to the longitudinal direction of the upper block **20**. When the upper block **20** is fixed to the lower block **10** by the bolt **60**, the upper block **20** bends slightly to generate elastic force, and the sealing performance of the gasket **41** can be improved.

[0043] Next, a joint structure according to a second embodiment will be described. The same components as those of the joint structure **1** of the first embodiment are denoted by the same reference numerals and will not be described.

[0044] FIG. **7** shows a cross-sectional view of a joint structure **101** according to the second embodiment.

[0045] As shown in FIG. **7**, the joint structure **101** includes the lower block **10**, the upper block **120**, the sleeve **30**, the gasket portion **40**, the pin **50**, and the bolt **60**.

[0046] The upper block **120** has a substantially rectangular parallelepiped shape and is disposed on the upper surface **12** of the lower block **10**. The upper block **120** is formed with a third pin insertion slit **121**, a third sleeve insertion hole **122**, and a third bolt insertion hole **123**. In a longitudinal direction of the upper block **120**, the third sleeve insertion hole **122** is located at the center, and the third pin insertion slit **121** and the third bolt insertion hole **123** are located on both sides of the third sleeve insertion hole **122**. The third pin insertion slit **121** opens to an upper surface **124** and a lower surface **125** of the upper block **120**, and also opens to one side surface **126** of the upper block **120** in the longitudinal direction. The third pin insertion slit **121** has a semi-oval shape (semi-elongated circular shape or semi-elliptical shape) in plan view, similar to the first pin insertion slit **21** (FIG. **2A**).

[0047] The third sleeve insertion hole **122** opens to the upper surface **124** and the lower surface **125** of the upper block **120**. The lower surface **125** of the upper block **120** has a fourth annular recess **128** formed around the opening of the third sleeve insertion hole **122**. The third bolt insertion hole **123** opens to the upper surface **124** and the lower surface **125** of the upper block **120**. The third sleeve insertion hole **122** and the third bolt insertion hole **123** have a substantially circular shape in plan view. The upper part of the upper block **120**, which is a portion forming the third pin insertion slit **121**, is provided with a pin counterbore portion **121A**. The upper part of the upper block **120**, which is a portion forming the third bolt insertion hole **123**, is provided with a bolt counterbore portion **121B**. The first head **51** of the pin **50** abuts against the pin counterbore portion **121A**, and the second head **61** of the bolt **60** abuts against the bolt counterbore portion **121B**.

[0048] The lower surface **125** of the upper block **120** is formed by a pair of inclined surfaces **177**. The pair of inclined surfaces **177** makes the upper block **120** thickest at its central portion in the longitudinal direction and thinnest at its ends. That is, the upper block **120** has the pair of inclined surfaces **177** that gradually extends away from the upper surface **124** from both ends in the longitudinal direction toward the center. The upper block **120** has a downward protrusion **178** formed by the pair of inclined surfaces **177**.

[0049] When the assembly of the joint structure **101** is completed, the upper surface **124** of the upper block **120** and the upper surface **12** of the lower block **10** are parallel. The dimension from the upper surface **12** of the lower block **10** to the lower surface **51A** of the first head **51** of the pin **50** (from the step surface **55** to the lower surface **51A**) is configured to be larger than the thickness from an upper surface of the pin counterbore **121A** of the pin step portion **121A** to the end of the lower surface **125** on the third pin insertion slit **121** side. Therefore, the upper block **120** can be inserted between the upper surface **12** of the lower block **10** and the lower surface **51A** of the first head **51** of the pin **50** while the upper block **120** is tilted, and the gasket **41** can be deformed to obtain a desired sealing effect. The inclination angle of each inclined surface **177** is preferably between 7° and 25° .

[0050] According to the above joint structure **101**, the third pin insertion slit **121** of the upper block **120** opens to one side surface **126** of the upper block **120** in the longitudinal direction, and the

lower surface **125** of the upper block **120** is formed by a pair of inclined surfaces **177** that gradually extends away from the upper surface **124** of the upper block **120** from both ends in the longitudinal direction toward the center. With this configuration, the upper block **120** and the sleeve **30** can be inserted between the first head **51** of the pin **50** and the lower block **10** from the lateral direction in a slightly inclined state, and the first shaft portion **52** of the pin **50** can be inserted into the third pin insertion slit **121**. This makes it easy to assemble the upper block **120** provided with the sleeve **30** to the lower block **10**.

[0051] Note that the present disclosure is not limited to the above-mentioned embodiment. A person skilled in the art can make various additions, modifications, and the like within the scope of the present disclosure.

[0052] For example, in the above embodiment, the angle of the inclined surface **77** was 5° , but it may be any angle between 5° and 15° . The inclination angles of the pair of inclined surfaces **77** and **177** were the same, but the angles may be changed. The lower surface **75** of the auxiliary member **70** is configured as an inclined surface **77**, but the lower surface of the auxiliary member **70** may be a flat surface and the upper surface of the upper block **20** may be configured as an inclined surface.

Claims

1. A joint structure comprising: an upper block arranged on a lower block having a fluid passage, in which a first pin insertion slit, a first sleeve insertion hole, and a first bolt insertion hole are formed, and the first sleeve insertion hole is located between the first pin insertion slit and the first bolt insertion hole; an auxiliary member provided on an upper surface of the upper block, in which a second pin insertion slit, a second sleeve insertion hole, and a second bolt insertion hole are formed, and the second sleeve insertion hole is located between the second pin insertion slit and the second bolt insertion hole; a sleeve inserted into the first sleeve insertion hole of the upper block and the second sleeve insertion hole of the auxiliary member, an inside of which serves as a fluid passage communicating with the fluid passage of the lower block; a pin having a first head, a first shaft portion, and a first male threaded portion, the first head abutting against an upper surface of the auxiliary member, the first shaft portion being inserted into the first pin insertion slit and the second pin insertion slit, and the first male threaded portion being screwed into the lower block; and a bolt having a second head, a second shaft portion, and a second male threaded portion, the second head abutting against the upper surface of the auxiliary member, the second shaft portion being inserted into the first bolt insertion hole and the second bolt insertion hole, and the second male threaded portion being screwed into the lower block, wherein the first pin insertion slit opens to one side surface of the upper block in a longitudinal direction, the second pin insertion slit opens to one side surface of the auxiliary member in the longitudinal direction, and a lower surface of the auxiliary member is formed by a pair of inclined surfaces that gradually extends away from the upper surface of the auxiliary member from both ends in the longitudinal direction toward a center.
2. The joint structure according to claim 1, wherein the first pin insertion slit and the second pin insertion slit have a semi-oval shape in plan view, and the second sleeve insertion hole and the second bolt insertion hole have an oval shape in plan view.
3. The joint structure according to claim 1, wherein the upper surface of the upper block is formed with a first notch that extends from the first sleeve insertion hole to penetrate a side surface parallel to the longitudinal direction of the upper block.
4. The joint structure according to claim 1, wherein the lower surface of the upper block is formed with a second notch that extends from the first sleeve insertion hole to penetrate a side surface parallel to the longitudinal direction of the upper block.
5. A joint structure comprising: an upper block arranged on a lower block having a fluid passage, in which a third pin insertion slit, a third sleeve insertion hole, and a third bolt insertion hole are formed, and the third sleeve insertion hole is located between the third pin insertion slit and the

third bolt insertion hole; a sleeve inserted into the third sleeve insertion hole of the upper block, the inside of which serves as a fluid passage communicating with the fluid passage of the lower block; a pin having a first head, a first shaft portion, and a first male threaded portion, the first head abutting against an upper side of the upper block, the first shaft portion being inserted into the third pin insertion slit, and the first male threaded portion being screwed into the lower block; and a bolt having a second head, a second shaft portion, and a second male threaded portion, the second head abutting against the upper side of the upper block, the second shaft portion being inserted into the third bolt insertion hole, and the second male threaded portion being screwed into the lower block, wherein the third pin insertion slit opens to one side surface of the upper block in a longitudinal direction, and a lower surface of the upper block is formed by a pair of inclined surfaces that gradually extends away from the upper surface of the upper block from both ends in the longitudinal direction toward a center.

6. The joint structure according to claim 5, wherein the third pin insertion slit have a semi-oval shape in plan view.
